

Elena Callus

Postdoctoral researcher

Date of birth: 17th July, 1994

Nationalities: Maltese & Finnish

Languages: Maltese, English, Finnish

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Topics of interest: quantum information theory, quantum computing with continuous-variables, quantum resource theories

Date of CV: September 18, 2026

Research positions

Sep 2026 – present **WINQ fellow**

Nordita, Stockholm, Sweden

Working as an independent researcher at Nordita ([institute webpage](#)), with a focus on resource theories and quantum information theory in CV systems.

Sep 2023 – Aug 2026 **Postdoctoral researcher**

Friedrich–Schiller–Universität (FSU), Jena, Germany

Working in the research group of Prof. Martin Gärttner at the Institute of Condensed Matter Theory and Optics ([group webpage](#)), with a focus on entanglement in CV systems with a phase-space approach.

Oct 2019 – Jul 2023 **Doctoral researcher**

University of Sheffield, United Kingdom

Working in the theory branch of the LDSD group ([group webpage](#)), researching quantum optics. In particular, investigating applications of light–matter interaction in entanglement generation and syndrome extraction for quantum error correction, as well as unidirectional photonic transport in waveguide-coupled nanocavities with experimental collaboration.

Education

Oct 2019 – Jul 2023 **Doctoral candidate (PhD)**

University of Sheffield, United Kingdom

Supervised by Prof. Pieter Kok, working in the LDS group within the Department of Physics & Astronomy. Thesis: *Applications of light-matter interaction in semiconductor structures* ([link to thesis](#)). Defence held on 13th September 2023.

Oct 2014 – Jul 2018 **Bachelor of Science (Honours)**

University of Malta, Malta

Joint honours degree in physics & mathematics. Graduated with first class honours (*summa cum laude*). Bachelor dissertation supervised by Dr André Xuereb: *A quantum perspective on random walks on graphs*.

Publications

1. **E. Callus**, M. Gärttner, T. Haas. *Revealing entanglement through local features of phase-space distributions*. *arXiv*: 2602.21688 (2026).
2. **E. Callus**, M. Gärttner, T. Haas. *Interferometric detection of continuous-variable entanglement using two states*. *Phys. Rev. A* **112**, 033704 (2025).
3. N. J. Martin, D. Hallett, M. Duda, L. Hallacy, **E. Callus**, L. Brunswick, R. Dost, E. Clarke, P. K. Patil, P. Kok, M. S. Skolnick, L. R. Wilson. *Purcell-Enhanced, Directional Light-Matter Interaction in a Waveguide-Coupled Nanocavity*. *Optica* **12** (7), 1100-1108 (2025).
4. **E. Callus**, P. Kok. *Spin-augmented observables for efficient photonic quantum error correction*. *Phys. Rev. A* **108**, 042613 (2023).
5. **E. Callus**, P. Kok. *Cumulative generation of maximal entanglement between spectrally distinct qubits using squeezed light*. *Phys. Rev. A* **104**, 052407 (2021).

Supervision & teaching

- **Co-supervision** of bachelor's thesis of M. Schubert (FSU Jena, 2026). *Calculating Concentratable Entanglement of Interesting Classes of States*.

- **Co-supervision** of master's thesis of V. Menhofer (FSU Jena, 2026).
Multipartite entanglement in loop-based architectures.
 - **Co-supervision** of bachelor's thesis of B. Bützler (FSU Jena, 2025).
Entanglement and multimode squeezing in high harmonic generation.
 - **Supervision** of master's research internship of F. Armbruster (FSU Jena, 2025).
Separability criteria in continuous-variable systems.
 - **Supervision** of summer internship of M. Duda (University of Sheffield, 2022).
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- **Co-ordinating & teaching** of master's module "Quantum information theory" (FSU Jena, 2025/26).
 - **Tutor** for "Fundamentals of Modern optics" & "Pre-course in Mathematics" (master's modules), "Optics & waves" (bachelor's module).

Service to the scientific community

- **Referee** for following international scientific journals: *Physical Review Letters*, *Physical Review A*, *Physical Review B*, *Physical Review Research*, *AVS Quantum Science*.
- **Mentor** for a 48-hour hackathon, "Quantum Ideas Factory 2026" at FSU Jena.